

Charges, Couplings, and Renormalization as Standing Transport

Charge-Response Carriers, Coupling Moduli, RG Ledgers, and No Primitive Standing from Numerical Interaction Strengths

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Abstract

This paper is SM-4 in the Standard Model carrier-exhaustion sequence. It applies the SM-1 master classifier to charge numbers, gauge couplings, running parameters, renormalization-group transport, matching conditions, calibration values, effective-field-theory coefficients, and unification displays after receiving the SM-2 gauge-sector closure and SM-3 matter-sector closure. The target is not numerical derivation of g_1, g_2, g_3, e, α , not proof of gauge-coupling unification, not calculation of beta functions, not exclusion of BSM model-building, and not the global SM-6 endpoint. The target is scoped primitive-status closure for the charge/coupling/renormalization arena.

The paper distinguishes primitive charge-response standing from derivative numerical and transport displays. Charge response, charge-witness standing, and the Standard-Model charge lattice belong to the primitive charge-side carrier package. Coupling values, running curves, bare parameters, RG crossings, scheme coordinates, threshold matches, calibration conventions, EFT coefficients, and unification plots belong to registered nonprimitive transport, calibration, bookkeeping, projection, or burden statuses unless a same-scope primitive carrier certificate closes. The positive coupling result is a standing-preserving coupling-envelope theorem, not numerical coupling derivation.

The charge/coupling/renormalization closure package is the ordered pair

$$\mathcal{K}_{CCR} = (C_{CCR}^{\text{prim}}, T_{CCR}^{\text{reg}}),$$

where

$$C_{CCR}^{\text{prim}} = \{C_{\text{charge-response}}, C_{\text{charge-witness}}, C_{\text{charge-lattice}}\}$$

and

$$T_{CCR}^{\text{reg}} = \{T_{\text{coupling-modulus}}, T_{\text{standing-preserving-coupling}}, T_{\text{RG-transport}}, T_{\text{scheme/matching}}, T_{\text{calibration}}, T_{\text{EW/EM-bookkeeping}}, T_{\text{EFT/projection}}\}.$$

The first component is the primitive charge-side carrier package. The second component is the registered status package for coupling, RG, scheme, matching, calibration, EW/EM bookkeeping, and EFT/projection claims. Using the SM-1 classifier, SM-2 gauge handoff, SM-3 matter handoff, Abelian charge-witness normalization, charge-ratio rigidity, Standard-Model charge-lattice rigidity, hypercharge/electric-charge quantization, electroweak/electromagnetic coupling bookkeeping, gauge-coupling forced uniqueness, dimensionless coupling moduli, renormalization

as standing-preserving scale transport, parameter run/non-run classification, physical-constant fixity, physical-constant role classification, and the Standing-Preserving Coupling Theorem, the paper proves

$$\text{Survprim}_{D_{\text{CCR}}}(\text{Attr}(D_{\text{CCR}}) \setminus C_{\text{CCR}}^{\text{prim}}) = 0.$$

This is a scoped charge/coupling/RG primitive-status theorem. It preserves ordinary charge, coupling, renormalization, EFT, matching, calibration, and BSM practice while deleting primitive standing from numerical values, coordinate displays, running plots, and model names alone.

The AASC/kernel dependency used here is not an optional interpretive overlay on charge, coupling, or renormalization physics. It is the minimal non-degeneracy condition under which charge/coupling/RG claims preserve standing, reference, sameness, comparison, prediction, and falsification. A coupling value, running curve, or calibration datum may be physically real and evidentially decisive without thereby becoming an independent primitive carrier.

Keywords: AASC; Standard Model; charge response; charge lattice; gauge couplings; renormalization group; RG transport; coupling moduli; physical constants; hypercharge; electric charge; EFT; matching; scheme dependence; calibration ledger; primitive closure; derivative skin; BSM audit.

Contents

1	Kernel Status: Non-Degeneracy, Not Overlay	5
	Reading roadmap	6
2	Reader orientation: what SM-4 proves	6
2.1	One-sentence thesis	6
2.2	SM-4 in the sequence	6
2.3	Standard usage preserved	6
2.4	Primitive-status deletion	7
3	Dependency ledger and proof-class boundary	7
3.1	Upstream kernel dependency and source direction	7
3.2	Imported SM-1 machinery	7
3.3	Imported SM-2 and SM-3 machinery	7
3.4	CCR dependency package	8
3.5	Closure proof spine	9
3.6	Theorem-status ladder	9
4	The charge/coupling/renormalization arena	9
5	CCR package and certificate structure	10
6	CCR carrier and status register	12
7	Charge response, charge witness, and charge lattice	12
7.1	Charge-response carrier	12
7.2	Charge-witness and charge-lattice carriers	12
8	Electroweak/electromagnetic coupling bookkeeping	13

9	Coupling values as moduli	14
9.1	Coupling-value claims	14
10	Standing-Preserving Coupling Theorem	15
11	Renormalization as standing-preserving scale transport	16
12	Scheme, matching, thresholds, and EFT	16
13	Parameter run/non-run classification	17
14	Calibration and measurement witnesses	18
15	BSM, unification, and EFT branch certificates	18
15.1	Worked hostile branch: unification plot or RG crossing	18
15.2	Extra Abelian factors, dark photon, and kinetic mixing	19
15.3	Worked hostile branch: dark photon or kinetic mixing	19
15.4	Fixed point, asymptotic safety, and EFT anomaly	20
15.5	Fine-tuning, anthropic, and multiverse variation	20
16	CCR skin atlas	20
17	CCR prediction box	20
17.1	Target-preserving prediction and falsification	20
17.2	Sector prediction statement	21
18	CCR Coverage Theorem	21
19	Final scoped CCR closure	22
19.1	Primitive charge-carrier closure	22
19.2	Registered CCR status closure	23
19.3	CCR package closure	23
19.4	Certificate-to-deletion lemma	23
20	Minimal hostile CCR reconstruction	24
20.1	Kernel-preserved hostile reconstruction	24
21	Referee objection lock	25
22	Standard physics orientation	25
23	Lean 4 formalization target and public audit reference	26
24	Handoff to later papers	28
25	Conclusion	28
A	Certificate ledger	28
B	BSM branch matrix	30

C	Theorem inventory	30
D	Source inventory	31
E	SM-6 release certificate	32
F	Detailed proof-sketch ledger	33
G	Full audit certificates for common CCR claims	34
G.1	Gauge coupling value audit form	34
G.2	Running coupling audit form	35
G.3	Unification plot audit form	35
G.4	EFT coefficient audit form	35
H	CCR branch algorithm	36
I	Evidence severity tiers for CCR claims	36
J	Denial obligations for CCR counterexamples	37
K	Publication-facing summary	37

1 Kernel Status: Non-Degeneracy, Not Overlay

The AASC kernel is used here in its upstream role. It is not an interpretive overlay placed on physics, and it is not a discretionary taxonomy imposed after the physical content is fixed. It states the minimal non-degeneracy conditions under which a physical claim can retain standing, reference, sameness, comparison, prediction, and falsification at all. Without such conditions, the expressions “same claim”, “same target”, “same regime”, “same comparison”, “same prediction”, and “same failure” lose determinate force.

For this reason, SM-4 is not a vocabulary substitution for ordinary physics. It is a local CCR primitive-status closure theorem over kernel-preserved claimhood. The familiar objects in the CCR arena may be physically real, measurable, predictive, and indispensable in calculation. That does not by itself make them primitive carriers. Primitive standing requires a same-scope role that remains standing-bearing under the admissible transformations of the fixed target.

Definition 1.1 (Non-degenerate CCR claim). *A charge/coupling/renormalization claim is non-degenerate only when its target, reference-bearing status, admissible comparison class, and standing conditions are preserved. In notation,*

$$\text{NonDegenerateClaim}_{\text{CCR}}(x) \Rightarrow \text{KernelPreserved}_{\text{CCR}}(x).$$

The implication is not a convention of this paper; it records the necessity condition for the claim to remain an identifiable object of assertion, prediction, comparison, or failure.

Theorem 1.2 (Sameness requires kernel preservation). *For SM-4 claims in D_{CCR} , sameness is kernel-preserved sameness:*

$$\text{SameClaim}_{\text{CCR}}(x, y) \Rightarrow \text{KernelPreserved}_{\text{CCR}}(x, y).$$

Consequently,

$$\neg \text{KernelPreserved}_{\text{CCR}}(x, y) \Rightarrow \neg \text{SameClaim}_{\text{CCR}}(x, y).$$

A non-preserved target, comparison class, admissibility rule, or reference-bearing status yields a replacement, degeneration, target update, or failed comparison rather than the same claim under test.

Theorem 1.3 (Prediction and falsification require non-degenerate reference). *Every prediction or falsification claim in the CCR arena presupposes preserved target identity, comparison class, and reference standing:*

$$\begin{aligned} & \text{Prediction}_{\text{CCR}}(x) \vee \text{Falsification}_{\text{CCR}}(x) \\ & \Rightarrow \text{TargetPreserved}(x) \wedge \text{ComparisonClassPreserved}(x) \wedge \text{ReferenceStanding}(x). \end{aligned}$$

A prediction is a prediction of the same claim only under sameness preservation. A falsification is a falsification of the same claim only under target-preserving failure.

Proposition 1.4 (Kernel-denial burden). *A denial of the kernel is not a successful rejection of SM-4 unless it recovers the functions that the kernel supplies. Formally, any denial must satisfy the burden*

$$\begin{aligned} \text{KernelDenied}(x) \Rightarrow & \text{RecoverSameness}(x) \wedge \text{RecoverStanding}(x) \\ & \wedge \text{RecoverReference}(x) \wedge \text{RecoverFalsification}(x). \end{aligned}$$

Thus a CCR counterexample must preserve the same target, same comparison class, and same reference-bearing object while showing that charge-response, charge-witness, charge-lattice, coupling-modulus, calibration, and scale-transport statuses can be given without the kernel conditions used here.

Remark 1.5 (Proof class). *The result is not a Lagrangian reconstruction, not a numerical parameter derivation, not an empirical-finality theorem, and not a no-new-physics theorem. It is a same-scope primitive-status exhaustion theorem for the stated arena. New measurements, anomalies, effective descriptions, and target-changing discoveries remain admissible; they do not become unlicensed same-scope primitive carriers merely by being real or useful.*

Reading roadmap

SM-4 proceeds in six stages. Sections 2–4 fix the claim boundary, sequence role, dependency ledger, theorem status, and charge/coupling/renormalization arena. Sections 5–7 define the CCR classifier package, primitive charge carriers, charge-witness standing, charge-lattice closure, and charge-number non-promotion. Sections 8–13 treat coupling values, electroweak/electromagnetic bookkeeping, coupling moduli, SPCT, RG transport, scheme/matching discipline, and parameter run/non-run classification. Sections 14–17 classify calibration values, EFT coefficients, BSM/unification branches, fixed-point claims, and varying-constant claims. Sections 18–20 give the CCR Coverage Theorem, package closure, scoped deletion theorem, and hostile reconstruction. The appendices provide ledgers, certificates, branch matrices, prediction register, formalization targets, and SM-6 handoff.

2 Reader orientation: what SM-4 proves

2.1 One-sentence thesis

Charges and couplings are not primitive because they are numbers; charge standing is carrier-indexed, while coupling values, RG flows, schemes, matching conditions, and EFT coefficients are standing-preserving transport, calibration, or projection ledgers.

This claim does not demote charge and coupling language. It preserves the ordinary physics while blocking an illicit inference: the availability of a numerical value, running curve, scheme coordinate, or unification display does not by itself supply primitive standing.

2.2 SM-4 in the sequence

SM-1 constructed the classifier and no-promotion discipline [2]. SM-2 closed the gauge-sector carrier package and handed forward the gauge-coupling non-carrier boundary [3]. SM-3 closed the matter-sector carrier package and handed forward mass/Yukawa non-promotion and matter-route discipline [4]. SM-4 closes charge/coupling/RG primitive-status escape. The local output handed to SM-6 is the closure package $\mathcal{K}_{\text{CCR}} = (C_{\text{CCR}}^{\text{prim}}, T_{\text{CCR}}^{\text{reg}})$ and the scoped primitive theorem

$$\text{Survprim}_{D_{\text{CCR}}}(\text{Attr}(D_{\text{CCR}}) \setminus C_{\text{CCR}}^{\text{prim}}) = 0.$$

2.3 Standard usage preserved

SM-4 does not deny electric charge, hypercharge, charge quantization, gauge couplings, running couplings, beta functions, RG flow, renormalization schemes, matching conditions, threshold corrections, EFT coefficients, calibration measurements, unification plots, fixed points, universality classes, or BSM coupling structures. These are lawful languages. The question is whether they carry primitive standing or a registered downstream status.

2.4 Primitive-status deletion

The denied inference is

$$\begin{aligned} & \text{charge number / coupling value / RG plot / scheme coordinate} \\ & \Rightarrow \text{primitive Standard Model standing.} \end{aligned}$$

The positive claim is

$$\begin{aligned} & \text{charge/coupling/RG standing} \\ & \Rightarrow \text{charge carrier or registered standing-transport status.} \end{aligned}$$

Nonclaim 2.1 (What SM-4 does not claim). *SM-4 does not derive numerical g_1, g_2, g_3, e, α , the weak mixing angle, exact beta functions, or numerical running values. It does not prove gauge coupling unification, prohibit fixed points, prohibit coupling anomalies, adjudicate every named BSM model, predict EFT coefficients, or close the global Standard Model endpoint. It closes primitive promotion by CCR skins under the fixed CCR arena.*

3 Dependency ledger and proof-class boundary

3.1 Upstream kernel dependency and source direction

The dependency order is fixed as follows. The kernel/admissibility layer supplies the minimal non-degeneracy conditions for sameness, standing, reference, comparison, prediction, and falsification. SM-4 instantiates those conditions in the CCR arena; it does not derive the kernel from the Standard Model or from any later structural-spine paper. Later Standard Model structural results may instantiate the same kernel at larger scope, but they do not supply the upstream necessity of kernel preservation.

Accordingly, dependence on the kernel is not a defect in the proof class. A successful objection must either break a received local closure premise, supply a faithful same-scope primitive certificate, or recover non-degenerate physical claimhood without the kernel functions listed in Proposition 1.

3.2 Imported SM-1 machinery

SM-4 uses the SM-1 component normal form

$$\begin{aligned} R_x = (D_{\text{SM}}, x, C_x, \text{Role}_x, \text{Rep}_x, \text{Meas}_x, \text{Transport}_x, \text{Route}_x, \\ \text{Mix}_x, \text{Proj}_x, \text{Extension}_x, \text{Update}_x, \text{Cert}_x, \rho_x, \text{Pred}_x). \end{aligned}$$

The imported results are carrier admission, skin identification, no primitive skin promotion, target-update separation, counterexample burden as review state, non-ad-hoc classifiability, prediction-register format, and falsification discipline [2].

3.3 Imported SM-2 and SM-3 machinery

SM-2 supplies charge-response standing, the gauge-sector carrier package, the gauge-coupling non-carrier boundary, EW/gauge interface discipline, and the BSM gauge branch posture [3]. SM-3 supplies matter-response closure, charged and neutral matter role discipline, mass/Yukawa non-promotion, fixed family-cardinality boundary, and detector/route/projection status discipline [4]. SM-4 uses these results as inputs and does not reprove them.

3.4 CCR dependency package

The CCR proof uses the source layers in Table 1. Each source is used only for the output listed. Numerical coupling closure is not imported from any source.

Table 1: SM-4 dependency ledger

Layer	Imported output	Use in SM-4	Nonclaim boundary
SM-1 classifier	Carrier/skin grammar, target update, burden, prediction status	Global classifier for CCR claims	Does not close CCR sector alone
SM-2 gauge carriers	Charge-response carrier, gauge package, coupling non-carrier boundary	Positive charge carrier and gauge handoff	Does not derive couplings
SM-3 matter carriers	Matter-response closure, mass/Yukawa non-promotion	Blocks re-promotion through matter sector	Does not close CCR sector alone
Abelian Charge-Witness Normalization	Abelian charge coordinates as witness-preserving refinements; normalization skin discipline	Charge witness and normalization non-promotion	Does not derive full SM charge lattice alone
Abelian Charge-Ratio Rigidity	Charge-ratio survivor discipline and no arbitrary Abelian coordinate proliferation	Rigidity for charge-ratio standing	Does not derive gauge couplings
Standard-Model Charge Lattice	Fixed-scope charge-lattice reconstruction and terminal quotient	Positive support for $C_{\text{charge-lattice}}$	Does not derive coupling values
L1 Hypercharge/Electric Charge	Hypercharge normalization and electric charge quantization under admissible redescription	SM-facing electric charge status	Fixed chiral content boundary
Gauge-Coupling Forced Uniqueness	Raw numeric gauge-coupling values as chart-level quantities; uniqueness at overlap/status level	Coupling value non-primitivity and forced compatibility	Not numerical coupling derivation
Dimensionless Couplings as Moduli	Dimensionless gauge couplings as moduli with determinability limits	Modulus status and non-derivability theorem	Does not select values
Renormalization Transport	RG flow as standing-preserving scale transport; scheme, threshold, EFT discipline	Positive RG transport ledger	Does not compute beta functions

Layer	Imported output	Use in SM-4	Nonclaim boundary
Why Some Parameters Run	Parameter mobility theorem with five mobility modes	Run/non-run classification	Does not fix numerical values
Standing Fixity of Constants	Standing-bearing constants are not freely variable same-domain parameters	Fixed-domain root for variation claims	Compatible with running and EFT
Physical Constants Classification	Five-role taxonomy for constant-like quantities	Role classification for couplings and constants	Not a numerical constants derivation
EW/EM Coupling Bookkeeping	Electroweak and electromagnetic coupling bookkeeping collapse	$T_{EW/EM}$ -bookkeeping and no overcount of g_1, g_2, e, θ_W	Does not derive values
SPCT	Standing-preserving coupling envelope G_{SP}	Positive coupling-envelope status	Not a generative primitive theorem

3.5 Closure proof spine

The proof architecture is

$$\begin{aligned}
& SM1 + SM2 + SM3 + PSM4 + SR4 + Q_{SM} \\
& + L1_{charge} + L1_{EW/EM} \\
& + CouplingModuli + CouplingUniqueness + SPCT \\
& + RGTransport + RunNonRun + ConstantFixity + ConstantClassification \\
& \Rightarrow \text{ClosedCCRPackage}(\mathcal{K}_{CCR}).
\end{aligned}$$

The final deletion theorem follows only after the CCR Coverage Theorem and hostile-branch discipline are stated. Package closure is not a device for absorbing counterexamples; it is a compact record of individual package closure, coverage, and exposed hostile branch conditions.

3.6 Theorem-status ladder

Definitions fix the CCR arena and statuses. Criteria govern carrier/status admission. Imported theorems are used only at their declared output. Closed SM-4 theorems are local CCR primitive-status results. Sector-local corollaries are handed to SM-5 and SM-6. Open burdens are live CCR challenges without closed certificates. The global Standard Model endpoint remains SM-6.

4 The charge/coupling/renormalization arena

Definition 4.1 (CCR arena). *The charge/coupling/renormalization arena is*

$$\begin{aligned}
D_{CCR} = (D_{SM}, C_{gauge}, C_{matter}, Q_{SM}, ChWit_{SM}, Lat_Q, G_{SP}, RGAdm, \\
SchemeAdm, MatchAdm, EFTAdm, ConstClass, CCRAdm).
\end{aligned}$$

It is the fixed arena in which charge numbers, coupling values, RG curves, matching data, calibration values, EFT coefficients, and charge/coupling BSM claims are evaluated.

Definition 4.2 (CCR claim). *CCRClaim(x) means that x is a submitted claim about a charge number, hypercharge value, electric charge, charge normalization, charge lattice, gauge coupling, bare coupling, measured coupling, running coupling, beta function, RG trajectory, fixed point, threshold, matching scale, scheme coordinate, renormalization convention, EFT coefficient, unification plot, coupling anomaly, varying constant, fine-tuning claim, anthropic or multiverse coupling claim, or BSM charge/coupling extension.*

Definition 4.3 (Same-scope CCR preservation). *SameScope_{D_{CCR}}(x) holds when x preserves D_{SM} , the gauge carrier package, matter carrier package, charge-witness ledger, charge lattice, anomaly/vacuum ledger, physical-constant role taxonomy, standing-preserving coupling envelope, scale-transport discipline, source/measurement domain, no-hidden-load condition, regime, and admissibility boundary.*

Criterion 4.4 (CCR target update). *A CCR claim is a target update only if it changes carrier inventory, charge lattice, gauge-product inventory, matter representation inventory, anomaly/vacuum ledger, source state, physical-constant role taxonomy, admissibility boundary, hidden-sector carrier inventory, EFT regime in a target-changing way, or the comparison arena.*

5 CCR package and certificate structure

Definition 5.1 (Two-level CCR closure package). *The CCR closure package is the ordered pair*

$$\mathcal{K}_{\text{CCR}} = (C_{\text{CCR}}^{\text{prim}}, T_{\text{CCR}}^{\text{reg}}),$$

where

$$C_{\text{CCR}}^{\text{prim}} = \{C_{\text{charge-response}}, C_{\text{charge-witness}}, C_{\text{charge-lattice}}\}$$

is the primitive charge-side carrier package and

$$T_{\text{CCR}}^{\text{reg}} = \{T_{\text{coupling-modulus}}, T_{\text{standing-preserving-coupling}}, T_{\text{RG-transport}}, T_{\text{scheme/matching}}, T_{\text{calibration}}, T_{\text{EW/EM-bookkeeping}}, T_{\text{EFT/projection}}\}$$

is the registered nonprimitive status package. Thus $\mathcal{K}_{\text{CCR}}$ is not a primitive carrier set. It records the primitive charge carriers together with the registered transport, calibration, bookkeeping, and projection ledgers into which coupling, RG, scheme, matching, calibration, and EFT claims descend.

Definition 5.2 (Primitive carrier certificate). *For a primitive charge-side CCR carrier C_i ,*

$$\begin{aligned} \text{ClosedCarrierCert}_{\text{CCR}}(C_i) := & \text{RoleCl}(C_i) \wedge \text{RedInv}(C_i) \wedge \text{Lic}(C_i) \\ & \wedge \text{NoDesc}(C_i) \wedge \text{NoSel}(C_i) \wedge \text{CarrierClose}(C_i). \end{aligned}$$

This certificate is used only for entries intended to carry primitive charge-side standing.

Definition 5.3 (Registered status certificate). *For a registered nonprimitive CCR status ledger T_j ,*

$$\begin{aligned} \text{ClosedStatusCert}_{\text{CCR}}(T_j) := & \text{StatusRole}(T_j) \wedge \text{RedInv}(T_j) \wedge \text{Lic}(T_j) \\ & \wedge \text{NonCarrier}(T_j) \wedge \text{NoSel}(T_j) \wedge \text{StatusClose}(T_j). \end{aligned}$$

A closed status certificate does not make T_j primitive. It closes the nonprimitive destination into which same-scope CCR claims descend.

Definition 5.4 (Candidate primitive CCR certificate). *For a submitted object x that asserts primitive CCR standing, $\text{CCRCert}(x)$ abbreviates the candidate primitive carrier-certificate obligation. In the final hostile branch this obligation must close as $\text{ClosedCarrierCert}_{\text{CCR}}(x)$, not as a registered status certificate.*

Definition 5.5 (Hostile branch and nonclosure guard).

$$\begin{aligned} \text{HostileCertificateBranch}_{\text{CCR}}(x) := & \text{SameScope}_{D_{\text{CCR}}}(x) \wedge \text{Primitive}_{\text{CCR}}(x) \\ & \wedge x \not\Downarrow C_{\text{CCR}}^{\text{prim}} \wedge x \not\Downarrow T_{\text{CCR}}^{\text{reg}} \\ & \wedge \text{NonDerivative}(x) \wedge \text{ClosedCarrierCert}_{\text{CCR}}(x). \end{aligned}$$

The hostile branch is not a registered status. It is the exposed defeat branch. Define

$$\text{NoClosedHostileCCR}(D_{\text{CCR}}, \mathcal{K}_{\text{CCR}}) := \neg \exists x \text{HostileCertificateBranch}_{\text{CCR}}(x).$$

If such a branch closes, SM-4 is falsified rather than saved by reclassification.

Definition 5.6 (Package closure).

$$\begin{aligned} \text{ClosedCCRPackage}(\mathcal{K}_{\text{CCR}}) := & \left(\bigwedge_{C_i \in C_{\text{CCR}}^{\text{prim}}} \text{ClosedCarrierCert}_{\text{CCR}}(C_i) \right) \\ & \wedge \left(\bigwedge_{T_j \in T_{\text{CCR}}^{\text{reg}}} \text{ClosedStatusCert}_{\text{CCR}}(T_j) \right) \\ & \wedge \text{Coverage}_{D_{\text{CCR}}}(\mathcal{K}_{\text{CCR}}) \wedge \text{NoClosedHostileCCR}(D_{\text{CCR}}, \mathcal{K}_{\text{CCR}}). \end{aligned}$$

Package closure is not a device for absorbing counterexamples. It is the conjunction of primitive carrier closure, registered status closure, coverage, and absence of a closed same-scope hostile primitive outside the package.

Lemma 5.7 (CCR skin identification). *A CCR object is skin-eligible or registered-status-eligible when its standing depends on a numerical value, normalization choice, scheme choice, scale choice, calibration route, measurement route, matching convention, EFT basis, RG parameterization, unification display, fixed-point chart, BSM model name, target-changing extension, or failure of invariant role preservation.*

Theorem 5.8 (CCR skin non-promotion).

$$\text{CCRSkin}(s) \wedge \neg \text{ClosedCarrierCert}_{\text{CCR}}(s) \Rightarrow \neg \text{Primitive}_{\text{CCR}}(s).$$

In particular, charge numbers, coupling values, running curves, scheme coordinates, matching conditions, calibration values, EFT coefficients, fixed-point displays, and unification plots do not self-promote into primitive CCR standing.

Proof. By the SM-1 no-promotion theorem, a submitted object that depends on a display, coordinate, route, projection, measurement, scheme, model surface, or downstream value lacks primitive standing unless an independent primitive carrier certificate closes. The CCR skin-identification lemma supplies the local dependence tests. Therefore a CCR skin or registered-status object without a closed primitive carrier certificate is not primitive. If a future object supplies such a certificate, it enters the carrier test or hostile certificate branch; it is not promoted by the skin itself. $\square$

6 CCR carrier and status register

Table 2: CCR carrier and registered-status package

Entry	Type	Role	Licensed language	Serious falsifier
Charge-response	primitive carrier	Charge-response standing received from SM-2	electric charge displays, currents, detector charge readouts	same-scope charge-response primitive outside carrier
Charge-witness	primitive carrier	Boundary-detectable charge witness	raw charge labels, weights, current labels	charge witness outside response discipline
Charge-lattice	primitive carrier	Fixed-scope SM charge lattice and terminal quotient	fractional labels, hypercharge displays, electric-charge units	same-scope non-lattice charge standing
Coupling modulus	registered status	Dimensionless gauge couplings as moduli	g_i, e, α , bare and measured couplings	numerical value generates primitive standing
SP coupling	registered status	Standing-preserving coupling envelope G_{SP}	effective couplings, interaction-strength parameters	coupling generates carrier identity
RG transport	registered transport	Scale-transport ledger	running curves, beta functions, fixed-point displays	RG curve generates primitive standing
Scheme/matching	registered transport	Scheme and threshold matching ledger	matching scales, renormalization schemes	scheme coordinate supplies primitive standing
Calibration	registered witness	Empirical calibration and measurement ledger	extracted constants, uncertainty intervals	measured value generates primitive carrier
EW/EM bookkeeping	registered bookkeeping	EW/EM coupling equivalence	g_1, g_2, e, θ_W displays	bookkeeping coordinate supplies independent primitive tensor load
EFT projection	registered projection	Low-energy EFT projection and operator basis status	Wilson coefficients, EFT anomalies, low-energy fits	EFT coefficient supplies primitive standing

7 Charge response, charge witness, and charge lattice

7.1 Charge-response carrier

$C_{\text{charge-response}}$ is received from SM-2. It is not a numerical charge value; it is standing-bearing response to charge-relevant gauge structure. A detector charge readout or a current label may witness this role, but the witness does not create the carrier.

7.2 Charge-witness and charge-lattice carriers

$C_{\text{charge-witness}}$ is the boundary-detectable charge-witness ledger supplied by Abelian charge-witness normalization. $C_{\text{charge-lattice}}$ is the fixed-scope Standard-Model charge-lattice survivor. Together they license discrete charge classes and electric-charge quantization while blocking arbitrary raw normalization. Fractional-looking quark charges are lattice points, not primitive exceptions to charge-lattice standing.

Theorem 7.1 (Charge number non-promotion).

$$\text{ChargeNumber}(q) \wedge \neg \text{ClosedCarrierCert}_{\text{CCR}}(q) \Rightarrow \neg \text{Primitive}_{\text{CCR}}(q).$$

A charge number may be a witness, selector, normalization display, or lattice coordinate. It is not primitive standing by numerical label.

Proof. A raw charge number is not invariantly meaningful until it is tied to a charge-response and charge-witness ledger. Abelian normalization and charge-ratio rigidity identify which charge coordinates survive as witness-preserving status. The number alone is normalization-sensitive and representation-attached; absent $\text{ClosedCarrierCert}_{\text{CCR}}(q)$, it remains a licensed coordinate or failed primitive promotion. $\square$

Theorem 7.2 (Hypercharge and electric-charge status). *For any hypercharge or electric-charge claim x ,*

$$\begin{aligned} \text{ChargeClaim}_{Y/Q}(x) \Rightarrow x \downarrow C_{\text{charge-witness}} \vee x \downarrow C_{\text{charge-lattice}} \\ \vee \text{RegisteredStatus}(x) \vee \text{Burden}(x) \vee \text{FailPrim}(x). \end{aligned}$$

Proof. L1 hypercharge/electric-charge completion and charge-lattice rigidity place admissible charge assignments into the charge-witness and charge-lattice ledgers. Normalization choices, electric-charge units, and coordinate labels are licensed statuses. A claim that supplies no witness or lattice role and asserts primitive standing by label alone fails; a same-scope nonreducing primitive claim remains burdened until a certificate closes. $\square$

Theorem 7.3 (Normalization skin theorem).

$$\text{NormalizationChoice}(n) \wedge \neg \text{ClosedCarrierCert}_{\text{CCR}}(n) \Rightarrow \neg \text{Primitive}_{\text{CCR}}(n).$$

Normalization is licensed when it preserves the charge-witness and charge-lattice ledger. It is not primitive by convention alone.

8 Electroweak/electromagnetic coupling bookkeeping

Definition 8.1 (EW/EM bookkeeping status). $T_{\text{EW/EM-bookkeeping}}$ is the registered equivalence-status ledger relating e , g_1 , g_2 , and θ_W across electroweak and electromagnetic skins. The familiar relation

$$e = g_2 \sin \theta_W = g_1 \cos \theta_W$$

is bookkeeping equivalence across regimes, not independent primitive tensor load for every coordinate. Here g_1 denotes the hypercharge-coupling coordinate under the declared Standard-Model normalization convention. Alternative notations such as g' or GUT-normalized g_1 are normalization/bookkeeping skins unless a separate carrier certificate closes. The theorem does not depend on a raw naming convention for the Abelian coupling coordinate.

Theorem 8.2 (EW/EM overcount deletion). *A claim treating g_1 , g_2 , e , and θ_W as independently primitive low-energy electromagnetic tensor load is not admissible under fixed D_{CCR} . Such a claim receives*

$$T_{\text{EW/EM-bookkeeping}} \vee \text{Transport}(x) \vee \text{Calibration}(x) \vee \text{Burden}(x) \vee \text{FailPrim}(x),$$

unless a same-scope hostile certificate closes outside the CCR closure package.

Proof. Electroweak bookkeeping exposes coordinates adapted to the gauge-phase skin, while low-energy electromagnetic bookkeeping exposes a single interaction strength. Cross-skin transport identifies the shared invariant witness rather than multiplying primitive degrees of freedom. Treating each coordinate as independent primitive load either overcounts, imports a representative selector, or changes the target. The lawful status is registered EW/EM bookkeeping, transport, calibration, burden, or failure. $\square$

Theorem 8.3 (EW/EM coordinate-choice non-primitivity).

$$EWEMCoordinateChoice(c) \wedge \neg \text{ClosedCarrierCert}_{CCR}(c) \Rightarrow \neg \text{Primitive}_{CCR}(c).$$

The coordinate name or normalization convention used for the Abelian coupling does not create primitive CCR standing.

Nonclaim 8.4. *This theorem does not derive e , g_1 , g_2 , or θ_W numerically. It closes the primitive-status overcount route.*

9 Coupling values as moduli

9.1 Coupling-value claims

Coupling-value claims include claims about g_1, g_2, g_3 , e , α , bare couplings, measured couplings, running couplings, threshold values, unified values, and dimensionless constants. SM-4 treats these as submitted surfaces, not as primitive standing certificates.

Definition 9.1 (Coupling-modulus status). $T_{\text{coupling-modulus}}(g)$ means that g is a coordinate or modulus in the admissible comparison space. A modulus may be real, measured, transport-relevant, and indispensable without being a primitive carrier.

Definition 9.2 (Negative modulus status and positive coupling-envelope status). $T_{\text{coupling-modulus}}$ is the negative and non-derivability status of dimensionless coupling values: it records that a coupling coordinate is a lawful modulus rather than a structurally selected primitive number. $T_{\text{standing-preserving-coupling}}$ is the positive standing-preserving coupling-envelope status supplied by SPCT. The two statuses are complementary rather than competing: modulus status blocks numerical primitive promotion, while SPCT licenses coupling language as an envelope constraint on already admitted standing.

Proposition 9.3 (Coupling-modulus/SPCT ordering).

$$\begin{aligned} \text{CouplingValue}(g) \Rightarrow & T_{\text{coupling-modulus}}(g) \vee T_{\text{standing-preserving-coupling}}(g) \vee T_{\text{calibration}}(g) \\ & \vee T_{\text{RG-transport}}(g) \vee \text{Burden}(g) \vee \text{FailPrim}(g). \end{aligned}$$

$T_{\text{standing-preserving-coupling}}$ does not promote g into a primitive carrier. It licenses g as an envelope constraint on standing-preserving interaction structure already admitted by gauge and matter carriers.

Theorem 9.4 (Coupling value non-primitivity).

$$\text{CouplingValue}(g) \wedge \neg \text{ClosedCarrierCert}_{CCR}(g) \Rightarrow \neg \text{Primitive}_{CCR}(g).$$

Proof. Gauge-coupling values vary under admissible reparameterization, scheme, calibration, scale transport, or normalization convention. They can witness or calibrate interaction strength, but their numerical display is not invariant carrier standing. If g supplies no certificate beyond value availability, it descends to $T_{\text{coupling-modulus}}$, $T_{\text{calibration}}$, $T_{\text{RG-transport}}$, or failure/burden status. $\square$

Theorem 9.5 (Dimensionless coupling non-derivability).

$$\text{DimensionlessCoupling}(g) \Rightarrow \text{ModulusStatus}(g) \vee \text{Calibration}(g) \vee \text{Transport}(g) \\ \vee \text{Burden}(g) \vee \text{TargetUpdate}(g).$$

Dimensionless status does not imply structural derivability.

Proof. The dimensionless-coupling-moduli result separates structural admissibility from numerical determinability. Symmetry, locality, anomaly cancellation, and admissibility can constrain the comparison arena without selecting a unique numerical coordinate. A claim that fixes a number must supply empirical, boundary, dynamical, or target-changing structure. Without that, the value is a modulus, calibration value, transport coordinate, burden, or target update. $\square$

Proposition 9.6 (Forced uniqueness boundary). *Gauge-coupling forced uniqueness is overlap/status uniqueness, not numerical uniqueness:*

$$\text{AdmissibleOverlap}(g_i, g_j) \Rightarrow \text{ForcedStatusCompatibility}(g_i, g_j).$$

The false replacement

$$g_i = \text{derived number}$$

is not a theorem of SM-4.

10 Standing-Preserving Coupling Theorem

Definition 10.1 (Standing-preserving coupling envelope). G_{SP} is the envelope of effective coupling values for which the relevant interaction-defining tokens preserve standing inside the declared admissible family. The registered status $T_{\text{standing-preserving-coupling}}$ records membership, transport, or boundary relation to that envelope.

Theorem 10.2 (SPCT status theorem).

$$\text{CplClaim}(g) \wedge \text{SameScope}_{D_{\text{CCR}}}(g) \Rightarrow g \downarrow T_{\text{standing-preserving-coupling}} \vee \text{Transport}(g) \vee \text{Calibration}(g) \\ \vee \text{Burden}(g) \vee \text{FailPrim}(g).$$

Proof. SPCT gives a positive coupling role: couplings can delimit the standing-preserving interaction envelope. This is not a generative primitive role. If a coupling claim preserves the target, it either records envelope membership, scale transport, calibration, or an unresolved certificate. If the primitive assertion rests on the coupling number alone, it fails primitive promotion. $\square$

Corollary 10.3 (Coupling non-generation).

$$T_{\text{standing-preserving-coupling}}(g) \Rightarrow \neg \text{GeneratesPrimitiveCarrier}(g).$$

Couplings constrain standing-preserving interaction envelopes. They do not create the primitive objects or carrier roles whose interactions they parameterize.

11 Renormalization as standing-preserving scale transport

Definition 11.1 (RG transport status). $T_{\text{RG-transport}}$ is standing-preserving transport across admissible scale descriptions. A running curve is a ledger of scale transport, not primitive standing.

Theorem 11.2 (RG transport theorem).

$$\begin{aligned} RGClaim(r) \wedge \text{SameScope}_{D_{\text{CCR}}}(r) \Rightarrow r \downarrow T_{\text{RG-transport}} \vee \text{Scheme}(r) \vee \text{Matching}(r) \\ \vee \text{Projection}(r) \vee \text{Burden}(r) \vee \text{FailPrim}(r). \end{aligned}$$

Proof. RG maps connect scale-indexed descriptions by preserving overlap witnesses, factoring through standing quotients, and respecting effective envelopes. The standing-bearing content is not raw numerical sameness of $g(\mu)$, but the transport class through which observables and matching data persist across scale. Therefore RG language descends to $T_{\text{RG-transport}}$, scheme/matching, projection, burden, or failure. $\square$

Theorem 11.3 (RG non-generation).

$$RGCurve(\gamma) \wedge \neg \text{ClosedCarrierCert}_{\text{CCR}}(\gamma) \Rightarrow \neg \text{Primitive}_{\text{CCR}}(\gamma).$$

Proof. A curve is a representation of transport. It may display flow, threshold behavior, fixed-point approach, or scheme dependence. None of those displays supplies primitive standing unless an independent CCR certificate closes. A same-scope primitive survivor must defeat transport, scheme, projection, and calibration descent. $\square$

Proposition 11.4 (Fixed point and universality status).

$$\begin{aligned} FixedPoint(f) \Rightarrow \text{Transport}(f) \vee \text{QuotientStatus}(f) \vee \text{UniversalityStatus}(f) \\ \vee \text{Burden}(f) \vee \text{TargetUpdate}(f). \end{aligned}$$

A fixed point can be a powerful structural or evidential status. It is not primitive carrier standing by the name “fixed point” or by chart display alone.

12 Scheme, matching, thresholds, and EFT

Theorem 12.1 (Scheme coordinate non-primitivity).

$$\text{SchemeCoordinate}(s) \wedge \neg \text{ClosedCarrierCert}_{\text{CCR}}(s) \Rightarrow \neg \text{Primitive}_{\text{CCR}}(s).$$

A scheme is a coordinate choice for transport/calibration. It does not carry primitive standing.

Theorem 12.2 (Matching condition status).

$$\begin{aligned} MatchingCondition(m) \Rightarrow T_{\text{scheme/matching}}(m) \vee \text{Projection}(m) \\ \vee \text{Burden}(m) \vee \text{TargetUpdate}(m). \end{aligned}$$

Threshold matching is a boundary/interface ledger, not primitive standing by condition alone.

Theorem 12.3 (EFT coefficient status).

$$\begin{aligned} EFTCoefficient(c) \Rightarrow T_{\text{EFT/projection}}(c) \vee \text{Transport}(c) \vee \text{Calibration}(c) \\ \vee \text{Burden}(c) \vee \text{TargetUpdate}(c). \end{aligned}$$

If c lacks a CCR certificate, then $\neg \text{Primitive}_{\text{CCR}}(c)$.

Proof. EFT coefficients are basis, matching, scheme, scale, and projection dependent unless a carrier-pressure certificate is supplied. Their evidential value is preserved: a coefficient can witness a mismatch, raise a burden, or motivate a target-update test. But coefficient availability does not by itself generate primitive CCR standing. $\square$

13 Parameter run/non-run classification

Definition 13.1 (Constant role taxonomy). *A constant-like or parameter-like object in a fixed physical arena has one of the role classes*

$$\text{ConstRole}(x) \in \{\text{AdmissibilityFixed}, \text{QuotientDependent}, \text{Effective}, \\ \text{TransportDefined}, \text{PresentationDependent}\},$$

unless it remains burdened or changes the target.

Theorem 13.2 (Fixity/running reconciliation).

$$\text{AdmissibilityFixed}(x) \Rightarrow \neg \text{Runs}_{\text{same-scope}}(x),$$

while

$$\text{Runs}(x) \Rightarrow \text{TransportDefined}(x) \vee \text{Effective}(x).$$

Consequently, a running coupling is not a contradiction of standing fixity. It is transport-defined or effective drift inside an admissible scale envelope; a standing-fixed role cannot be varied inside the same scope by coordinate choice.

Theorem 13.3 (Parameter mobility theorem).

$$\text{ParameterClaim}(p) \Rightarrow \text{AdmissibilityFixed}(p) \vee \text{QuotientDependent}(p) \vee \text{Effective}(p) \\ \vee \text{TransportDefined}(p) \vee \text{PresentationDependent}(p) \\ \vee \text{Burden}(p) \vee \text{TargetUpdate}(p).$$

Proof. The physical-constant classification theorem exhausts the legitimate roles played by constant-like quantities under fixed-domain admissibility. The run/non-run theorem then sorts apparent variation into presentation change, effective drift, transport-defined running, quotient-stable invariance, or admissibility-fixed non-runnability. No homogeneous parameter space is created merely by writing several quantities as numbers. $\square$

Theorem 13.4 (Fixity and running reconciliation).

$$\text{Runs}(p) \Rightarrow \text{TransportDefined}(p) \vee \text{Effective}(p),$$

and

$$\text{AdmissibilityFixed}(p) \Rightarrow \neg \text{Runs}_{\text{same-scope}}(p).$$

A structural anchor cannot run by coordinate choice inside the same admissibility domain. A running coupling is not a contradiction of standing fixity. It is transport-defined or effective drift inside an admissible scale envelope. A standing-fixed role cannot be varied within the same scope by coordinate choice; a transport-defined parameter can run because the running is the ledger of scale transport, not a primitive change in carrier identity.

Proposition 13.5 (Varying-constant branch). *A varying-constant claim receives*

$$\text{Transport}(x) \vee \text{EffectiveDrift}(x) \vee \text{PresentationChange}(x) \vee \text{Burden}(x) \vee \text{TargetUpdate}(x).$$

It is not automatically a same-scope primitive survivor.

14 Calibration and measurement witnesses

Definition 14.1 (Calibration status). $T_{\text{calibration}}(m)$ means that m is a measurement or calibration witness of an already typed charge/coupling/transport role.

Theorem 14.2 (Measurement non-generation).

$$\text{MeasuredCoupling}(m) \wedge \neg \text{ClosedCarrierCert}_{\text{CCR}}(m) \Rightarrow \neg \text{Primitive}_{\text{CCR}}(m).$$

A measured value can witness, constrain, calibrate, or raise burden severity. It does not create primitive standing.

Proof. Measurement status is relational: it connects an empirical readout to a carrier or transport role. If the measured numerical value generated primitive standing, calibration route would bypass the carrier and transport tests. That is exactly the primitive-promotion path deleted by the classifier. $\square$

Uncertainty intervals, fit procedures, detector calibrations, and extraction conventions remain lawful. They are measurement, calibration, transport, or projection statuses.

15 BSM, unification, and EFT branch certificates

Theorem 15.1 (CCR BSM branch theorem). For any same-scope CCR BSM claim b ,

$$\begin{aligned} & \text{CCRB SMClaim}(b) \wedge \text{SameScope}_{D_{\text{CCR}}}(b) \\ & \Rightarrow b \downarrow C_{\text{CCR}}^{\text{prim}} \vee b \downarrow T_{\text{CCR}}^{\text{reg}} \\ & \vee \text{RegisteredOtherStatus}_{\text{CCR}}(b) \\ & \vee \text{CounterexampleBurden}_{\text{CCR}}(b) \\ & \vee \text{FailPrim}_{\text{CCR}}(b) \\ & \vee \text{HostileCertificateBranch}_{\text{CCR}}(b). \end{aligned}$$

If the claim is not same-scope, then

$$\begin{aligned} & \text{CCRB SMClaim}(b) \wedge \neg \text{SameScope}_{D_{\text{CCR}}}(b) \\ & \Rightarrow \text{TargetUpdate}(b). \end{aligned}$$

Proof. A same-scope BSM claim remains a CCR claim and enters the SM-1 classifier. It either descends to primitive charge-side carrier standing, descends to a registered transport/calibration/bookkeeping/projection ledger, receives another registered status, remains burdened, fails primitive promotion by label/value/plot alone, or supplies a hostile carrier certificate. A target-changing claim is classified as target update rather than same-scope survivor. $\square$

15.1 Worked hostile branch: unification plot or RG crossing

A plotted or calculated crossing of running couplings is the natural pressure case for SM-4. The submitted claim is: a crossing of running couplings supplies primitive unification standing. The branch sequence is fixed.

1. **Target test.** If the claim changes D_{CCR} , gauge carrier inventory, charge lattice, matter representation inventory, anomaly/vacuum compatibility, physical-constant role taxonomy, or scale-transport discipline, it is TargetUpdate.

2. **Descent test.** If the crossing is scheme-dependent, threshold-dependent, matching-dependent, normalization-dependent, or scale-coordinate-dependent, it descends to $T_{\text{RG-transport}}$, $T_{\text{scheme/matching}}$, $T_{\text{calibration}}$, or $T_{\text{EFT/projection}}$.
3. **Projection/status test.** If the crossing is a model comparison, GUT display, or extrapolation, it receives projection, extension, or burden status.
4. **Burden test.** If same-scope primitive unification standing is asserted but the carrier certificate is not closed, the claim is `CounterexampleBurden`.
5. **Failure test.** If primitive standing is asserted from a plot, numerical coincidence, or unification group name alone, the claim is `FailPrim`.
6. **Hostile branch.** A genuine falsifier requires

$$\text{SameScope}_{D_{\text{CCR}}}(u) \wedge \text{ClosedCarrierCert}_{\text{CCR}}(u) \wedge u \not\Downarrow C_{\text{CCR}}^{\text{prim}} \wedge u \not\Downarrow T_{\text{CCR}}^{\text{reg}} \wedge \text{NonDerivative}(u).$$

A unification plot can be evidentially serious, but it is not primitive standing by plotted crossing alone. A same-scope primitive unification carrier with a closed certificate would falsify SM-4.

15.2 Extra Abelian factors, dark photon, and kinetic mixing

A new Abelian factor, dark-photon label, or kinetic-mixing term is not primitive by group label, charge label, or coupling value. It receives extension, projection, transport, burden, target update, or failed primitive status unless it preserves the charge lattice, anomaly/vacuum ledger, gauge/matter carrier inventory, and no-hidden-load condition while closing a hostile CCR carrier certificate.

15.3 Worked hostile branch: dark photon or kinetic mixing

A dark-photon or kinetic-mixing claim is the most common low-energy CCR pressure case after unification. The submitted claim is: a new vector label or kinetic-mixing coefficient supplies primitive electromagnetic or charge/coupling standing inside the fixed CCR arena. The branch sequence is fixed.

1. **Target test.** If the claim introduces hidden carrier inventory, a new source state, a changed charge lattice, a changed anomaly/vacuum ledger, or a changed comparison arena, it is `TargetUpdate`.
2. **Bookkeeping and projection test.** If the claim is a kinetic-mixing coordinate, field-basis choice, low-energy resonance surface, or detector topology, it descends to $T_{\text{EFT/projection}}$, $T_{\text{scheme/matching}}$, $T_{\text{calibration}}$, or projection status.
3. **Extension test.** If the claim preserves the primitive charge-side carriers while adding a carrier-preserving model surface or route, it is an extension or projection disposition, not a new primitive carrier.
4. **Burden test.** If same-scope primitive electromagnetic or charge/coupling standing is asserted outside $C_{\text{CCR}}^{\text{prim}}$ but no carrier certificate closes, the claim remains `CounterexampleBurden`.
5. **Failure test.** If primitive standing is asserted from the phrase “dark photon”, from a kinetic-mixing parameter, from a fitted coefficient, or from a detector excess alone, the claim is `FailPrim`.
6. **Hostile branch.** A genuine falsifier requires

$$\text{SameScope}_{D_{\text{CCR}}}(d) \wedge \text{ClosedCarrierCert}_{\text{CCR}}(d) \wedge d \not\Downarrow C_{\text{CCR}}^{\text{prim}} \wedge d \not\Downarrow T_{\text{CCR}}^{\text{reg}} \wedge \text{NonDerivative}(d).$$

A dark-photon claim can be empirically important and still nonprimitive in the CCR sense. It becomes a same-scope SM-4 falsifier only if it supplies independent primitive charge/coupling standing while preserving the fixed arena and defeating the registered projection, transport, calibration, and extension statuses.

15.4 Fixed point, asymptotic safety, and EFT anomaly

Fixed-point and asymptotic-safety claims are classified as transport, quotient, universality, burden, or target update. EFT anomalies are projection, transport, calibration, burden, or target-update candidates. Neither fixed-point display nor Wilson coefficient supplies primitive CCR standing by itself.

15.5 Fine-tuning, anthropic, and multiverse variation

Fine-tuning, anthropic, and multiverse coupling claims normally change source state, comparison measure, ensemble authority, selector, or target regime. If they preserve same scope, they must still descend to constant-role taxonomy, transport/effective status, burden, or failed primitive status. Raw value scanning is not primitive standing.

16 CCR skin atlas

Table 3: CCR skin and status atlas

Object	Expected status	Non-promotion rule
charge number	witness or lattice coordinate	number does not create charge carrier
hypercharge label	normalization/lattice status	label does not create primitive standing
electric charge value	charge-response witness	value does not create primitive carrier
gauge coupling g_i	coupling modulus, calibration, or transport	coupling value does not create carrier
fine-structure constant α	calibration/modulus/status	number does not ground primitive EM standing
bare coupling	scheme/regularization coordinate	bare value is not primitive
running coupling	RG transport coordinate	curve does not create primitive standing
beta function	transport law/display	formula is not carrier
fixed point	quotient/transport/universality status	fixed point does not self-ground
matching scale	threshold/interface ledger	matching condition is not primitive
scheme coordinate	presentation/transport coordinate	scheme is not primitive
unification plot	projection/transport display	crossing is not primitive
EFT coefficient	projection/transport status	coefficient is not carrier
measured value	calibration witness	measurement does not generate primitive
varying-constant claim	transport, effective, update, or burden	variation is not automatically same-scope primitive

17 CCR prediction box

17.1 Target-preserving prediction and falsification

The prediction register in SM-4 has reality contact only because the target and comparison class are preserved. It is not an internal bookkeeping list. A predicted measurement, anomaly, route, or extension pressure is assessed as a prediction of the same charge/coupling/renormalization claim only when the reference-bearing target in D_{CCR} remains fixed.

Proposition 17.1 (Reality-contact discipline for CCR predictions). *For every CCR prediction or falsification entry,*

$$\begin{aligned} & \text{Prediction}_{\text{CCR}}(x) \vee \text{Falsification}_{\text{CCR}}(x) \\ \Rightarrow & \text{KernelPreserved}_{\text{CCR}}(x) \wedge \text{TargetPreserved}(x) \wedge \text{ComparisonClassPreserved}(x). \end{aligned}$$

If the target changes, the result is a target update. If the target is preserved but the alleged primitive status does not close, the result is a registered status, burden, or failed primitive promotion rather than a same-scope primitive survivor.

17.2 Sector prediction statement

$$\text{FutureCCRClear}(y) \Rightarrow \exists \rho_{\text{CCR}}(y) \in \text{Status}_{\text{CCR}} \quad \text{or} \quad \text{Failure}(\text{CCR0}).$$

Failure is not hidden by ad hoc status expansion. It is recorded as pressure on the classifier.

Table 4: SM-4 CCR prediction register

ID	Claim type	Expected disposition	Falsifier
CCR0	future CCR claim	carrier, transport, projection, calibration, update, burden, or failure	real claim cannot be classified without grammar expansion
CCR1	charge number	charge witness/lattice coordinate or failure	charge number alone supplies primitive standing
CCR2	normalization	charge-lattice/witness status	normalization convention supplies primitive standing
CCR3	gauge coupling value	modulus, calibration, transport, burden, or failure	numerical value supplies primitive carrier
CCR4	EW/EM coupling relation	bookkeeping equivalence	independent primitive EW/EM coordinates in the same low-energy interaction
CCR5	RG running curve	standing-preserving scale transport	running curve generates primitive standing
CCR6	scheme/matching condition	scheme/matching ledger	scheme coordinate supplies primitive standing
CCR7	unification plot	transport, matching, projection, burden, update, or failure	same-scope primitive unification carrier closes
CCR8	fixed point	quotient, transport, universality, burden, or update	fixed point supplies primitive standing without certificate
CCR9	EFT coefficient	projection, transport, calibration, burden, or update	coefficient alone supplies primitive standing
CCR10	varying constant	transport, effective drift, presentation, burden, or update	same-scope primitive variation closes outside taxonomy
CCR11	BSM charge/coupling claim	extension, projection, update, burden, failure, or hostile branch	same-scope primitive CCR carrier closes

18 CCR Coverage Theorem

Definition 18.1 (Registered other CCR status).

$$\begin{aligned} \text{RegisteredOtherStatus}_{\text{CCR}}(x) := & \text{Rep}(x) \vee \text{Meas}(x) \vee \text{Route}(x) \vee \text{Projection}(x) \\ & \vee \text{Extension}(x) \vee \text{RegisteredStatus}_{\text{other}}(x). \end{aligned}$$

This category contains registered statuses not already named in $T_{\text{CCR}}^{\text{reg}}$. Transport, scheme, matching, calibration, EW/EM bookkeeping, and EFT projection are handled by the named ledgers in $T_{\text{CCR}}^{\text{reg}}$.

Theorem 18.2 (CCR Coverage Theorem).

$$\begin{aligned}
CCRClaim(x) \wedge \text{SameScope}_{D_{CCR}}(x) \Rightarrow & x \downarrow C_{CCR}^{\text{prim}} \\
& \vee x \downarrow T_{CCR}^{\text{reg}} \\
& \vee \text{RegisteredOtherStatus}_{CCR}(x) \\
& \vee \text{CounterexampleBurden}_{CCR}(x) \\
& \vee \text{FailPrim}_{CCR}(x) \\
& \vee \text{HostileCertificateBranch}_{CCR}(x).
\end{aligned}$$

Proof. The SM-1 classifier supplies the finite grammar. Primitive charge claims either descend to $C_{\text{charge-response}}$, $C_{\text{charge-witness}}$, or $C_{\text{charge-lattice}}$, or they lack a certificate and become burden/failure. Coupling-value claims descend to modulus, calibration, standing-preserving coupling, or RG transport status unless they supply an independent primitive carrier certificate. RG, scheme, matching, EW/EM bookkeeping, calibration, and EFT claims descend to the registered ledgers in T_{CCR}^{reg} or enter burden/failure. Within the same-scope hypothesis, BSM and unification claims descend, receive registered extension or projection status, remain burdened, fail primitive promotion, or enter the hostile certificate branch. Claims that do not preserve the same scope are handled separately by target-update separation. These cases exhaust same-scope CCR claims without adding an ad hoc status. $\square$

19 Final scoped CCR closure

19.1 Primitive charge-carrier closure

$$\text{ClosedCarrierPackage}(C_{CCR}^{\text{prim}}) := \bigwedge_{C_i \in C_{CCR}^{\text{prim}}} \text{ClosedCarrierCert}_{CCR}(C_i).$$

The primitive carrier entries are $C_{\text{charge-response}}$, $C_{\text{charge-witness}}$, and $C_{\text{charge-lattice}}$. Charge response is received from SM-2 as a primitive response carrier. Abelian charge-witness normalization supplies the admissible witness form; charge-ratio rigidity blocks arbitrary normalization selectors; Standard-Model charge-lattice rigidity and L1 hypercharge/electric-charge quantization close the SM-facing charge lattice. Thus the charge-side primitive carriers close while raw charge numbers and normalization conventions remain skins.

The proof load for $C_{\text{charge-response}}$ is role-preserving response: it is not the numeral assigned to a charged object, but the standing-bearing response role through which charge claims remain comparable under admissible redescription. The proof load for $C_{\text{charge-witness}}$ is boundary-detectable witness preservation: a proposed charge coordinate must preserve the same charge-witness ledger rather than merely name a weight, current, sign, or normalization. The proof load for $C_{\text{charge-lattice}}$ is fixed-scope quotient closure: the Standard-Model-facing charge lattice is the terminal charge quotient for the declared arena, so non-lattice interpolation, arbitrary fractional deformation, or normalization drift cannot be counted as primitive charge standing.

These three closures are carrier closures in the strict sense. They are the only primitive entries in the CCR package. All numerical charge displays remain licensed coordinates or witnesses of this charge-side carrier package, not additional primitive carriers.

Theorem 19.1 (Primitive charge-carrier package closure).

$$\text{ClosedCarrierPackage}(C_{CCR}^{\text{prim}}).$$

19.2 Registered CCR status closure

$$\text{ClosedStatusPackage}(T_{\text{CCR}}^{\text{reg}}) := \bigwedge_{T_j \in T_{\text{CCR}}^{\text{reg}}} \text{ClosedStatusCert}_{\text{CCR}}(T_j).$$

The registered status entries are $T_{\text{coupling-modulus}}$, $T_{\text{standing-preserving-coupling}}$, $T_{\text{RG-transport}}$, $T_{\text{scheme/matching}}$, $T_{\text{calibration}}$, $T_{\text{EW/EM-bookkeeping}}$, and $T_{\text{EFT/projection}}$. Coupling forced-uniqueness and dimensionless-moduli support close the negative side: a numerical coupling coordinate is not primitive standing and is not structurally derived as a value. SPCT closes the positive side: couplings are licensed as standing-preserving envelope constraints over already-admitted interaction standing. RG flow, schemes, thresholds, matching conditions, and EFT coefficients close as transport, interface, calibration, or projection ledgers rather than primitive carriers. EW/EM bookkeeping equivalence closes the cross-skin status of e, g_1, g_2, θ_W under declared normalization. Calibration values close as measurement witnesses of already typed roles.

The registered status package is therefore not a second primitive carrier network. $T_{\text{coupling-modulus}}$ records the non-derivability and coordinate-modulus status of dimensionless coupling values; $T_{\text{standing-preserving-coupling}}$ records the positive standing-preserving coupling envelope in which those values constrain already-admitted interaction structure. $T_{\text{RG-transport}}$ records scale transport rather than same-domain primitive change; $T_{\text{scheme/matching}}$ records changes of scheme, threshold, and matching interface; $T_{\text{calibration}}$ records measurement and extraction; $T_{\text{EW/EM-bookkeeping}}$ records cross-skin bookkeeping between electroweak and electromagnetic coupling coordinates; and $T_{\text{EFT/projection}}$ records projection and low-energy operator status.

Each T -entry closes as a nonprimitive destination. A closed status certificate says that same-scope claims of the relevant kind have a lawful registered status and therefore cannot survive as primitive standing by numerical, coordinate, or projection availability alone. It does not promote the registered ledger into a primitive carrier.

Theorem 19.2 (Registered CCR status package closure).

$$\text{ClosedStatusPackage}(T_{\text{CCR}}^{\text{reg}}).$$

19.3 CCR package closure

Theorem 19.3 (CCR package closure).

$$\text{ClosedCCRPackage}(\mathcal{K}_{\text{CCR}}).$$

Proof. The primitive charge-carrier package closes by the preceding carrier theorem. The registered nonprimitive status package closes by the preceding status theorem. The CCR Coverage Theorem covers every same-scope CCR claim: a claim descends to $C_{\text{CCR}}^{\text{prim}}$, descends to $T_{\text{CCR}}^{\text{reg}}$, receives another registered status, remains burdened, fails primitive promotion, or enters the hostile certificate branch. The nonclosure guard $\text{NoClosedHostileCCR}(D_{\text{CCR}}, \mathcal{K}_{\text{CCR}})$ is part of package closure. Therefore the CCR closure package is closed without promoting registered ledgers into primitive carriers. $\square$

19.4 Certificate-to-deletion lemma

Lemma 19.4 (Certificate-to-deletion).

$$\text{ClosedCCRPackage}(\mathcal{K}_{\text{CCR}}) \Rightarrow \text{Survprim}_{D_{\text{CCR}}}(\text{Attr}(D_{\text{CCR}}) \setminus C_{\text{CCR}}^{\text{prim}}) = 0.$$

Proof. Let $x \in \text{Attr}(D_{\text{CCR}}) \setminus C_{\text{CCR}}^{\text{prim}}$ be a same-scope primitive candidate. By coverage, x descends to $C_{\text{CCR}}^{\text{prim}}$, descends to $T_{\text{CCR}}^{\text{reg}}$, receives another registered status, is burdened, fails primitive promotion, or enters the hostile branch. The first case is impossible because x is outside $C_{\text{CCR}}^{\text{prim}}$. The $T_{\text{CCR}}^{\text{reg}}$ case is a registered nonprimitive status, not primitive carrier standing. Other registered statuses are derivative. Burden is not survivor, and failed primitive is not primitive. If x enters the hostile branch, NoClosedHostileCCR rules it out in the closed scope; if such a branch closes, SM-4 is falsified rather than proven. Therefore no same-scope primitive survivor remains outside $C_{\text{CCR}}^{\text{prim}}$. $\square$

Theorem 19.5 (Scoped CCR primitive-status deletion).

$$\text{Survprim}_{D_{\text{CCR}}}(\text{Attr}(D_{\text{CCR}}) \setminus C_{\text{CCR}}^{\text{prim}}) = 0.$$

Proof. The theorem follows from primitive charge-carrier closure, registered CCR status closure, CCR coverage, CCR skin non-promotion, target-update separation, burden non-survivor status, and the no-closed-hostile-branch guard. The result deletes only same-scope primitive standing outside the primitive charge-side carrier package. It does not delete valid charge values, coupling measurements, running curves, EFT coefficients, unification models, or target-updating discoveries. $\square$

20 Minimal hostile CCR reconstruction

20.1 Kernel-preserved hostile reconstruction

A hostile reconstruction has force only when it is a same-target reconstruction. It must preserve D_{CCR} , preserve the comparison class, preserve reference standing, and then supply an independent primitive role not already carried or licensed by the closed CCR package. Merely replacing the target, changing the admissibility rule, or appealing to a useful model surface is not a same-scope counterexample.

Criterion 20.1 (Faithful CCR counterexample). *A faithful CCR counterexample must show all of the following: (i) kernel-preserved same-scope membership in D_{CCR} ; (ii) primitive role occupancy; (iii) non-derivativeness from the certified CCR carriers; (iv) nonmembership in the registered status atlas; and (v) target-preserving failure of the no-promotion theorem. If any item fails, the claim is a registered status, burden, failed primitive, or target update rather than a primitive survivor.*

A real counterexample must be stronger than a new number, running curve, crossing plot, coefficient, or model name. Let y_{CCR}^* satisfy:

1. $\text{CCRClaim}(y_{\text{CCR}}^*)$;
2. $\text{SameScope}_{D_{\text{CCR}}}(y_{\text{CCR}}^*)$;
3. $y_{\text{CCR}}^* \not\downarrow C_{\text{CCR}}^{\text{prim}}$ and $y_{\text{CCR}}^* \not\downarrow T_{\text{CCR}}^{\text{reg}}$;
4. it is not a charge label, coupling value, RG curve, scheme coordinate, matching condition, calibration value, EFT coefficient, unification plot, fixed-point display, or model name;
5. it is not a target update;
6. it defeats derivative descent;
7. it supplies a closed primitive CCR carrier certificate outside $C_{\text{CCR}}^{\text{prim}}$.

Then

$$\text{Survprim}_{D_{\text{CCR}}}(y_{\text{CCR}}^* \setminus C_{\text{CCR}}^{\text{prim}})$$

and the local SM-4 theorem is falsified. The classifier may not relabel this object as target update unless the target-update criterion is satisfied, and it may not leave it as burden once the certificate closes.

21 Referee objection lock

Table 5: Referee objection lock

Objection	Response
Couplings are physical.	Granted. Their status is transport, calibration, modulus, or envelope status, not primitive standing by numerical value.
The paper does not derive g_i, e, α .	Correct. SM-4 explicitly denies numerical coupling derivation.
Running couplings are real.	Granted. Running is standing-preserving scale transport.
Unification plots are meaningful.	Granted. They are transport/projection evidence or burden/target-update candidates, not primitive carriers by plot alone.
Scheme dependence is technical.	Scheme dependence is precisely why scheme coordinates cannot be primitive standing.
EFT coefficients indicate new physics.	They can. Their status is projection, transport, burden, or target update unless primitive standing closes.
A fixed point is primitive.	Only if it supplies a same-scope carrier certificate outside the transport/quotient ledger.
This denies BSM.	BSM charge/coupling claims are audit-admissible; they are not protected from primitive-status classification.
AASC dependency is a defect.	It is the declared proof class. A standing objection must break a load-bearing theorem or supply a faithful same-scope counterexample.

22 Standard physics orientation

SM-4 preserves conventional use of charge, coupling, and renormalization language. Standard quantum field theory routinely distinguishes bare and renormalized parameters, scheme-dependent coordinates, measured couplings, beta-function displays, threshold matching, EFT Wilson coefficients, and low-energy effective descriptions. SM-4 does not replace that practice. It classifies the primitive-status role of those objects.

Charge labels and hypercharge assignments remain the ordinary way to display representation response and interaction neutrality. The paper’s claim is that charge standing is carried by charge-response, charge-witness, and charge-lattice roles, not by a raw number independently of those ledgers. Coupling constants remain indispensable empirical and transport quantities. The paper denies only that a coupling value, by being a number, is a primitive Standard Model carrier. RG flow remains the ordinary mechanism by which effective parameters vary with scale. The paper classifies that variation as standing-preserving transport rather than unlicensed same-domain primitive change. EFT coefficients and unification plots remain valuable discovery tools. They enter the classifier as projection, transport, burden, or target-update candidates unless a same-scope primitive certificate closes.

23 Lean 4 formalization target and public audit reference

The Lean 4 release is used here as a paper-shaped, proof-checked audit and certificate surface for the SM manuscript sequence. It records the endpoint carriers, declared hypotheses, theorem crosswalk inventory, and certificate interfaces in Lean; verifies that the stated closure theorems typecheck without `sorry`, `admit`, live local `axiom` declarations, or `unsafe` code in the audit surface; and ties the formal release to the DOI-bearing manuscript bundle. This is a substantive support layer: it gives a typed theorem registry, explicit hypothesis boundaries, machine-checked closure over declared certificate fields, an audit trail across SM-1–SM-6 and the endpoint paper, reproducibility checks, and DOI/release alignment. It should not be described as an independent reconstruction of the full Standard Model argument from foundational physics.

The generation-cardinality endpoint should be read at the governance-to-endpoint interface, not as a standalone phenomenological derivation inside the Standard Model sector. The mechanized surface represents the pre-physical admissibility constraints under which a non-degenerate physical presentation is admissible: open-arena discipline, no-selector and no-presentation-dependence constraints, role-occupancy closure, and boundary handoff. Within that framework, generation cardinality is a formal closure condition rather than a fit to particle content or an inference from arbitrary presentation-dependent structure. The Lean release audits that certificate surface and its dependencies; it does not reconstruct every analytic or phenomenological argument in the SM papers.

The public Lean-facing support package is the relevant SM ARC reference archive [21]: its README identifies the paper-shaped theorem surface, the locally received SM-1–SM-6 ARC layers, the generation-cardinality endpoint, the theorem-inventory layer, and the build/audit verifier. Its v0.1.0 release records the `GenerationCardinality` Lake target, carries SM-1 through SM-6 as local received ARC layers, preserves the generation-cardinality source artifacts, and reports build and focused axiom-audit checks over the endpoint anchors.

Lean audit item	Manuscript-facing role
Repository	somamaley-ux/AASC-Standard-Model-Carrier-Exhaustion-Generation-Cardinality, the public Lean 4 reference archive for the SM-1–SM-6 ARC carrier-exhaustion chain and the generation-cardinality endpoint.
Release	v0.1.0 - Standard Model Carrier Exhaustion / Generation Cardinality.
DOI	10.5281/zenodo.21219417.
Primary theorem surface	<code>SM4.lean</code> and its imported <code>SM4/...</code> files, read with <code>SM1.lean</code> – <code>SM3.lean</code> , <code>SM5.lean</code> , <code>SM6.lean</code> , and <code>GenerationCardinality.lean</code> as received support surfaces.
Axiom audit command	Repository verifier <code>scripts/check-generation-cardinality-audit.ps1</code> with the <code>-SkipLakeUpdate</code> option; the verifier builds the endpoint target, scans for live <code>axiom</code> , <code>sorry</code> , <code>admit</code> , and <code>unsafe</code> declarations, and runs focused axiom checks over endpoint anchors.
Build target	<code>GenerationCardinality</code> , with aggregate imports through <code>SM1.lean</code> – <code>SM6.lean</code> and <code>GenerationCardinality.lean</code> .

Audit meaning	The public release machine-checks the governance-to-endpoint certificate surface over declared carriers, received ARC layers, and explicit manuscript hypotheses. Once the pre-physical admissibility layer is accepted, the endpoint predicate is a formal closure condition, not a phenomenological fit or downstream presentation selector.
Certificate/interface value	The layer is not merely bookkeeping: it supplies a typed theorem registry, explicit hypothesis boundaries, machine-checked closure over declared certificate fields, an SM-1–SM-6 audit crosswalk, reproducibility checks, and DOI/release alignment.
Boundary	The cardinality endpoint is not a standalone SM-sector derivation from particle dynamics; it is a governance-level endpoint that makes explicit the admissibility, no-selector, open-arena, role-occupancy, and presentation-independence constraints under which later SM interpretations are meaningful. The Lean release audits this certificate/interface boundary and its dependencies; it does not reconstruct every analytic or phenomenological argument of SM-1–SM-6.

A manuscript-faithful Lean route for SM-4 represents:

1. fixed scopes and submitted-claim or candidate-record sets as explicit carriers;
2. carrier packages, registered-status packages, and received sector packages as typed structures rather than implicit background metaphysics;
3. same-scope and target-update predicates as explicit boundary conditions;
4. derivative-skin, representative, route, projection, transport, burden, failure, and extension statuses as typed predicates or classifiers;
5. certificate closure and no-promotion principles as theorems over declared hypotheses, not as global axioms;
6. finite branch exhaustions for the relevant same-scope arena;
7. the handoff to SM-6, or from SM-6 to the generation-cardinality endpoint, as an explicit typed support relation.

For SM-4, the paper-specific audit surface is the charge/coupling/RG status package: charge-response carrier, charge-witness and charge-lattice closure, coupling-modulus status, standing-preserving coupling transport, RG transport, scheme and matching status, calibration status, EW/EM bookkeeping collapse, EFT projection status, unification-plot non-promotion, CCR branch certificates, and scoped CCR primitive-status deletion.

A manuscript-faithful formalization over the declared carriers and hypotheses requires no custom axiom at a load-bearing node. If a proposed audit file relies on `axiom`, `sorry`, `admit`, `unsafe`, or an opaque proof-carrying placeholder at such a node, it is not counted as formal support for this manuscript. Conversely, a clean axiom audit is not a stand-alone validation of the physical hypotheses encoded as record fields; it is a proof-checked governance-to-endpoint certificate boundary whose value is typed traceability, explicit assumptions, reproducible build discipline, and DOI-aligned auditability.

24 Handoff to later papers

SM-4 hands to SM-5 the coupling and RG status boundaries for flavor-transition, CKM/PMNS transport, CP-phase coordinates, and basis/scheme dependence. It hands to SM-6 the closure package $\mathcal{K}_{CCR} = (C_{CCR}^{\text{prim}}, T_{CCR}^{\text{reg}})$, the charge/coupling/RG skin atlas, charge-witness and charge-lattice closure, coupling-modulus and SPCT status, RG/scheme/matching/EFT transport ledgers, EW/EM bookkeeping equivalence, CCR BSM/unification branch matrix, CCR prediction register, and the scoped deletion theorem

$$\text{Survprim}_{D_{CCR}}(\text{Attr}(D_{CCR}) \setminus C_{CCR}^{\text{prim}}) = 0.$$

SM-6 may then assemble

$$C_{SM}^{\text{prim}} = C_{\text{gauge}} \cup C_{\text{matter}} \cup C_{CCR}^{\text{prim}} \cup C_{\text{flavor}} \cup C_H \cup C_{\text{projection}},$$

with T_{CCR}^{reg} retained in the registered status atlas.

25 Conclusion

The kernel-status strengthening is part of the theorem, not a commentary layer. The closure claimed here is primitive-status closure under the minimal non-degeneracy conditions required for CCR claims to remain the same standing-bearing physical claims across measurement, prediction, comparison, and attempted falsification.

SM-4 closes the charge/coupling/renormalization primitive-status question under the declared same-scope CCR arena. Charge standing is carrier-indexed through charge response, charge witnesses, and the Standard-Model charge lattice. Coupling values, running curves, beta-function displays, matching thresholds, scheme choices, calibration values, EFT coefficients, fixed-point displays, and unification plots are registered transport, calibration, projection, extension, burden, target-update, failure, or hostile-certificate statuses. They are not primitive standing by numerical or coordinate availability alone. The registered ledgers in T_{CCR}^{reg} close their nonprimitive status; they are not included as primitive carriers. Coupling values are physically meaningful calibration, modulus, transport, and measurement statuses; SM-4 denies only that such values become primitive carriers by numerical availability alone.

Therefore,

$$\text{Survprim}_{D_{CCR}}(\text{Attr}(D_{CCR}) \setminus C_{CCR}^{\text{prim}}) = 0.$$

This is not empirical no-new-physics, numerical coupling derivation, gauge-coupling unification, or global Standard Model closure. It is the scoped local CCR deletion theorem handed forward to SM-6.

A Certificate ledger

Table 7: Carrier and status certificate discharge

Entry	Obligation	Discharge
$C_{\text{charge-response}}$	RoleCl	Received from SM-2 as charge-response carrier.
	RedInv	Stable under charge label, current display, detector readout, and normalization redescription.

Entry	Obligation	Discharge
$C_{\text{charge-witness}}$	Lic	Licenses electric-charge displays, current labels, and response witnesses.
	NoDesc/NoSel	No raw charge number or detector event selects primitive charge identity.
	CCRClose	Closes by SM-2 plus charge-witness/lattice support.
	RoleCl	Boundary-detectable charge witness from Abelian normalization.
	RedInv	Stable under Abelian coordinate and normalization changes preserving witness output.
$C_{\text{charge-lattice}}$	Lic	Licenses weights, characters, charge labels, and current displays.
	NoDesc/NoSel	No coordinate or normalization is primitive by itself.
	CCRClose	Closes by PSM4 and SR4.
	RoleCl	Fixed-scope SM charge lattice and terminal quotient.
	RedInv	Stable under global normalization and sign conventions.
$T_{\text{coupling-modulus}}$	Lic	Licenses familiar electric-charge lattice and hypercharge displays.
	NoDesc/NoSel	Non-lattice interpolation and arbitrary deformation fail or update target.
	CCRClose	Closes by charge-lattice rigidity and L1 charge support.
	RoleCl	Coupling value as modulus/status coordinate.
	RedInv	Stable as moduli status, not as raw numerical invariant.
$T_{\text{standing-preserving-coupling}}$	Lic	Licenses g_i, e, α , bare, measured, and running displays.
	NoDesc/NoSel	Numerical value does not select primitive carrier.
	CCRClose	Closes by forced uniqueness and dimensionless-moduli support.
	RoleCl	Standing-preserving coupling-envelope role G_{SP} .
	RedInv	Stable across admissible redescription families preserving interaction role.
$T_{\text{RG-transport}}$	Lic	Licenses effective coupling envelopes and standing thresholds.
	NoDesc/NoSel	Envelope membership does not generate primitive carrier identity.
	CCRClose	Closes by SPCT.
	RoleCl	Scale transport preserving standing quotients.
	RedInv	Stable across admissible RG maps and scheme changes at the transport level.
$T_{\text{scheme/matching}}$	Lic	Licenses running curves, beta functions, fixed-point displays.
	NoDesc/NoSel	Curves and fixed points do not self-ground primitive standing.
	CCRClose	Closes by RG transport and run/non-run support.
	RoleCl	Scheme and threshold interface ledger.
	RedInv	Stable as transport/interface status under coordinate changes.

Entry	Obligation	Discharge
$T_{\text{calibration}}$	Lic	Licenses schemes, regulators, matching scales, thresholds.
	NoDesc/NoSel	Scheme coordinates and thresholds are not primitive carriers.
	CCRClose	Closes by RG transport and constants taxonomy.
	RoleCl	Measurement and calibration witness status.
	RedInv	Stable as witness relation, not as primitive number.
	Lic	Licenses measured couplings, fitted values, uncertainty intervals.
$T_{\text{EW/EM-bookkeeping}}$	NoDesc/NoSel	Measurement does not create the primitive role.
	CCRClose	Closes by SM-1 measurement non-promotion and CCR typing.
	RoleCl	Bookkeeping equivalence across EW and EM skins.
	RedInv	Stable under admissible cross-skin transport.
	Lic	Licenses e, g_1, g_2, θ_W displays.
	NoDesc/NoSel	No coordinate receives independent primitive tensor load.
$T_{\text{EFT/projection}}$	CCRClose	Closes by EW/EM coupling bookkeeping completion.
	RoleCl	EFT projection and low-energy operator-basis status.
	RedInv	Stable as projection/matching status under basis changes.
	Lic	Licenses EFT coefficients and low-energy fits.
	NoDesc/NoSel	Coefficient alone is not primitive standing.
	CCRClose	Closes by RG/EFT transport discipline.

B BSM branch matrix

Table 8: CCR BSM branch matrix

Claim type	Same-scope branch	Target-update branch	Primitive falsifier branch
Unification plot	transport, projection, matching, or burden	larger-domain unification target	same-scope unification carrier outside $C_{\text{CCR}}^{\text{prim}}$ and $T_{\text{CCR}}^{\text{reg}}$
Extra $U(1)$	extension, burden, or failure if ledgers preserved	changed charge lattice or anomaly ledger	new Abelian CCR carrier preserving all ledgers
Dark photon	kinetic mixing/projection, extension, or burden	hidden-sector carrier inventory	independent EM/charge-coupling primitive outside package
Fixed point	quotient, transport, universality, or burden	changed UV target	fixed-point carrier outside transport ledger
EFT anomaly	projection, transport, calibration, or burden	changed effective regime or invariant bundle	coefficient pressure with closed primitive certificate
Varying constant	presentation, effective drift, transport, or burden	domain/source/measure change	same-scope primitive variation outside constant taxonomy
Fine tuning or multiverse	role classification or burden	ensemble or measure target change	same-scope selector authority with certificate closed

C Theorem inventory

Table 9: SM-4 theorem inventory

Theorem	Function	Status
CCR Skin Non-Promotion	Blocks primitive status from numeric/transport/projection skins	closed SM-4 theorem
Charge Number Non-Promotion	Blocks raw charge-number primitive standing	closed
Hypercharge/Electric-Charge Status	Routes charge claims to witness/lattice/status	closed
Normalization Skin Theorem	Blocks primitive status by normalization convention	closed
EW/EM Overcount Deletion	Collapses independent primitive load from EW/EM coordinates	closed
Coupling Value Non-Primitivity	Blocks coupling values as primitive carriers	closed
Dimensionless Coupling Non-Derivability	Marks coupling values as moduli/status without numerical derivation	closed
SPCT Status Theorem	Provides positive standing-preserving coupling-envelope status	closed
RG Transport Theorem	Classifies running as scale transport	closed
RG Non-Generation	Blocks running curves as primitive standing	closed
Scheme/Matching/EFT Status Theorems	Classify schemes, thresholds, EFT coefficients	closed
Parameter Mobility Theorem	Sorts run/non-run and constant-role claims	closed
CCR BSM Branch Theorem	Provides branch logic and hostile certificate branch	closed branch logic
CCR Coverage Theorem	Covers all same-scope CCR claims	closed
Scoped CCR Primitive-Status Deletion	Local SM-4 endpoint	closed

D Source inventory

Table 10: Source inventory used by SM-4

Title	Role in SM-4
Primitive Carriers, Derivative Skins, and Classifier-Level Prediction Closure in the Standard Model	SM-1 classifier, no-promotion theorem, target-update discipline, prediction register.
Gauge Carriers: Photon, Gluon, and Weak-Boson Standing	SM-2 gauge, charge-response, color, weak, anomaly, vacuum, and coupling-boundary handoff.

Title	Role in SM-4
Matter Carriers: Quarks, Leptons, and Neutrinos	SM-3 matter closure, mass/Yukawa boundary, detector/route/projection discipline.
Abelian Charge-Witness Normalization	PSM4 charge witness, normalization skin, and Abelian factor readiness.
Abelian Charge-Ratio Rigidity in the Admissible Gauge Arena	SR4 charge-ratio survivor discipline.
Rigidity of the Standard-Model Charge Lattice	Fixed-scope charge-lattice reconstruction.
L1 - Equivalence Completion (Paper 11): Hypercharge Normalization and Electric Charge Quantization under Admissible Redescription	Hypercharge/electric-charge normalization and quantization support.
Gauge-Coupling Parameters and Forced Uniqueness under Admissible Redescription	Raw numeric gauge-coupling non-primitivity and overlap-level forced compatibility.
Dimensionless Gauge Couplings as Moduli	Coupling moduli, structural non-derivability, determinability limits.
Renormalization as Standing-Preserving Scale Transport	RG flow, scheme, threshold, universality, EFT transport.
Why Some Parameters Run and Others Cannot	Parameter mobility and run/non-run classification.
The Standing Fixity of Physical Constants	Fixed-domain constraints on same-domain constant variation.
A Classification Theorem for Physical Constants	Role taxonomy for constants and parameters.
Equivalence Completion Between Electroweak and Electromagnetic Coupling Bookkeeping	EW/EM coupling bookkeeping collapse.
Standing-Preserving Coupling Theorem (SPCT)	Standing-preserving coupling envelope.

E SM-6 release certificate

SM-6 receives from SM-4:

1. the closed CCR closure package $\mathcal{K}_{\text{CCR}} = (C_{\text{CCR}}^{\text{prim}}, T_{\text{CCR}}^{\text{reg}})$;
2. the charge, coupling, RG, scheme, matching, calibration, EW/EM, and EFT skin atlas;
3. CCR BSM and unification branch certificates;
4. prediction register CCR0–CCR11;
5. the hostile CCR reconstruction;

6. the scoped primitive-status deletion theorem

$$\text{Survprim}_{D_{\text{CCR}}}(\text{Attr}(D_{\text{CCR}}) \setminus C_{\text{CCR}}^{\text{prim}}) = 0.$$

SM-6 must not convert this local theorem into a global endpoint without receiving the remaining flavor/mixing/generation and projection-interface closures.

F Detailed proof-sketch ledger

This appendix records the proof compression used in the main text. It is included so that each local theorem can be audited without confusing proof use with numerical phenomenology.

Table 11: Proof-sketch ledger for main SM-4 results

Result	Proof compression
Charge response	Received from SM-2 as a closed gauge-sector carrier. SM-4 does not identify charge response with a number; it uses charge response as the role that licenses charge witnesses and charge displays.
Charge witness	PSM4 supplies the passage from response witnesses to Abelian charge-coordinate language. The coordinate is admissible only when it preserves the boundary-detectable witness. A raw charge label that lacks this relation is skin or failure.
Charge lattice	The fixed-scope charge-lattice theorem shows that admissible SM-facing charge labels factor through a discrete lattice quotient. SM-4 uses this as charge-lattice standing; it does not treat a normalization convention as primitive.
EW/EM bookkeeping	The EW/EM equivalence paper shows that electroweak variables and low-energy electromagnetic coupling language overcount tensor load if treated as independent primitives for the same interaction. SM-4 records the result as bookkeeping status, not value derivation.
Coupling moduli	The coupling-moduli source shows that structural constraints do not select unique dimensionless gauge-coupling values without extra dynamical, boundary, or empirical structure. SM-4 uses this both negatively, to block primitive value standing, and positively, to license modulus status.
Forced uniqueness	Forced uniqueness holds at admissible-overlap and normalization status, not at raw numerical value. SM-4 uses this to prevent a false route from overlap compatibility to numerical derivation.
SPCT	SPCT supplies a standing-preserving envelope for effective coupling values. The envelope is a constraint on standing preservation and does not generate the objects or carrier roles coupled by the interaction.

Result	Proof compression
RG transport	RG flow is a lawful scale-transport relation preserving overlap witnesses and effective envelopes. The running coordinate is not the primitive standing object; the transport class is the registered status.
Scheme and matching	Scheme coordinates and matching thresholds coordinate descriptions across presentations or regimes. They become standing-relevant only through transport and interface preservation. They are not primitive by coordinate availability.
EFT projection	EFT coefficients are projection and matching coordinates. Their value can be evidentially serious, but it does not become primitive standing unless projection, measurement, transport, route, and target-update statuses are defeated.
Constant-role classification	The physical-constant role taxonomy prevents homogeneous-parameter ontology. Constants and couplings can be admissibility-fixed, quotient-dependent, effective, transport-defined, or presentation-dependent; writing them all as numbers does not put them in one primitive parameter space.

G Full audit certificates for common CCR claims

G.1 Gauge coupling value audit form

Field	Entry
Submitted object	A numerical value such as g_1 , g_2 , g_3 , e , or α .
Fixed target	D_{CCR} with gauge, matter, charge, constants, and transport ledgers fixed.
Carrier candidate	None by numerical value alone. The lawful positive status is modulus, calibration, standing-preserving coupling, or transport.
Skin tests	Value depends on normalization, scheme, calibration, scale, measurement route, or representative convention.
Descent route	$T_{\text{coupling-modulus}}$, $T_{\text{calibration}}$, $T_{\text{standing-preserving-coupling}}$, or $T_{\text{RG-transport}}$.
Target-update boundary	A claim changing carrier inventory, source state, or comparison regime is a target update.
Burden condition	Same-scope primitive numerical standing outside $C_{\text{CCR}}^{\text{prim}}$ and $T_{\text{CCR}}^{\text{reg}}$ is asserted but certificate remains open.
Terminal status	Modulus, calibration, transport, burden, update, or failed primitive.

G.2 Running coupling audit form

Field	Entry
Submitted object	A running curve $g(\mu)$, beta-function display, or RG trajectory.
Fixed target	D_{CCR} with scale-transport discipline fixed.
Carrier candidate	$T_{\text{RG-transport}}$, a registered scale-transport status.
Skin tests	Curve depends on scale coordinate, scheme, threshold convention, and presentation.
Descent route	RG transport, scheme/matching, EFT projection, calibration, or burden.
Target-update boundary	Changed UV target, hidden sector, or new carrier inventory is a target update.
Burden condition	A same-scope primitive RG object outside transport status is asserted but uncertified.
Terminal status	Transport, scheme, matching, projection, burden, update, or failure.

G.3 Unification plot audit form

Field	Entry
Submitted object	A plot, crossing, fitted unification scale, or unified coupling surface.
Fixed target	D_{CCR} with SM gauge and charge ledgers fixed.
Carrier candidate	None by plot alone.
Skin tests	Plot depends on RG coordinate, scheme, thresholds, matching convention, and projection.
Descent route	Transport, projection, matching, extension disposition, or burden.
Target-update boundary	A larger-domain unification theory changes the target if it changes carrier inventory or invariant bundle.
Burden condition	Same-scope primitive unification standing outside $C_{\text{CCR}}^{\text{prim}}$ and $T_{\text{CCR}}^{\text{reg}}$ is asserted without certificate.
Terminal status	Transport, projection, burden, target update, or failed primitive.

G.4 EFT coefficient audit form

Field	Entry
Submitted object	Wilson coefficient, low-energy operator coefficient, or EFT anomaly.
Fixed target	D_{CCR} with EFT/projection and matching discipline fixed.

Carrier candidate	$T_{\text{EFT/projection}}$ as projection status, not primitive carrier.
Skin tests	Operator basis, matching convention, renormalization scheme, and low-energy projection dependence.
Descent route	EFT projection, transport, calibration, burden, or target update.
Target-update boundary	A changed effective regime or new carrier inventory is target update.
Burden condition	Coefficient pressure defeats projection, transport, and measurement status but certificate remains open.
Terminal status	Projection, transport, calibration, burden, target update, or failure.

H CCR branch algorithm

A submitted CCR claim is evaluated in the following order:

1. Fix D_{CCR} and state the role bundle allegedly preserved.
2. Test target preservation: gauge carrier inventory, matter carrier inventory, charge lattice, anomaly/vacuum ledger, source state, physical-constant role taxonomy, and scale-transport discipline.
3. If target is changed, assign target update.
4. If same-scope is preserved, test whether a primitive charge carrier or standing-transport certificate closes.
5. If a primitive carrier certificate closes inside $C_{\text{CCR}}^{\text{prim}}$, assign carrier status; if a status certificate closes inside $T_{\text{CCR}}^{\text{reg}}$, assign the corresponding registered status.
6. If the claim descends to numerical value, charge label, coupling modulus, calibration, RG transport, scheme/matching, EFT projection, or bookkeeping status, assign registered status.
7. If same-scope primitive standing is asserted but the certificate is open, assign burden.
8. If primitive standing is asserted from value, plot, scheme, coefficient, or model label alone, assign failed primitive.
9. If a closed same-scope primitive certificate outside the CCR closure package is supplied, record the hostile certificate branch and mark SM-4 locally falsified.

I Evidence severity tiers for CCR claims

Table 16: Evidence tiers for charge/coupling/RG claims

Tier	Meaning	Effect on SM-4
0	new label, parameterization, scheme choice, charge convention, or model name	no endpoint pressure unless carrier standing is supplied
1	single measurement anomaly in a charge, coupling, or EFT coefficient	classify as measurement, calibration, transport, projection, burden, or update candidate

Tier	Meaning	Effect on SM-4
2	repeated coherent anomaly across experiments, scales, or routes	raises burden severity and motivates branch certificate or target-update test
3	carrier-pressure evidence resisting measurement, transport, scheme, matching, and projection classifications	serious burden; requires explicit CCR certificate search
4	same-scope primitive CCR survivor outside $C_{\text{CCR}}^{\text{prim}}$ and $T_{\text{CCR}}^{\text{reg}}$ with closed carrier certificate	falsifies local SM-4 primitive-status deletion theorem

J Denial obligations for CCR counterexamples

A CCR counterexample has standing only if it discharges the following obligations:

1. Target preservation: preserve D_{CCR} , not merely use charge or coupling vocabulary.
2. Carrier inventory preservation: do not add hidden sectors, new charge lattices, or changed source states while claiming same-scope status.
3. Derivative-status exclusion: show that the claim is not a charge label, coupling value, running curve, scheme coordinate, matching condition, calibration value, EFT coefficient, fixed-point display, or unification plot.
4. Descent defeat: defeat descent to $C_{\text{charge-response}}$, $C_{\text{charge-witness}}$, $C_{\text{charge-lattice}}$, $T_{\text{coupling-modulus}}$, $T_{\text{standing-preserving-coupling}}$, $T_{\text{RG-transport}}$, $T_{\text{scheme/matching}}$, $T_{\text{calibration}}$, $T_{\text{EW/EM-bookkeeping}}$, and $T_{\text{EFT/projection}}$, as relevant.
5. Certificate closure: supply role closure, redescription invariance, licensing proof, no-hidden-selector proof, descent exclusion, and local CCR closure.
6. Burden closure: move beyond live burden status into a closed certificate. Until then, the claim is neither accepted as primitive survivor nor dismissed.

K Publication-facing summary

Within the fixed Standard Model charge/coupling/renormalization comparison arena, no residual same-scope primitive attribute survives outside the primitive charge-side carrier package; coupling and RG claims are sorted into the registered standing-transport package. Charge numbers, hypercharge coordinates, electric-charge units, gauge-coupling values, running curves, beta-function displays, scheme choices, matching thresholds, calibration values, EFT coefficients, fixed-point displays, and unification plots remain lawful representatives, witnesses, transport ledgers, projection displays, extensions, target updates, burdens, failures, or genuine falsifiers according to the stated certificate conditions. SM-4 does not derive numerical coupling values or assert empirical finality. It closes primitive promotion by numerical, transport, calibration, and projection skins alone.

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